# Run R184 Method Validation Technical Note

HELIX FORGE LAB / p04-full-003370 / RUN R184 / 2026-07-13

| ALPHA F1 | PRECISION | RECALL | BETA F1 |
|---|---|---|---|
| 0.873 | 0.881 | 0.865 | 0.846 |

## Decision context

This report records a bounded method validation evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

## Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

## Lineage

Upstream evidence: p04-full-000197 (archive/closed/2026-07-13/run-r184/threshold-selection/specifications/run-r184-threshold-selection-metadata-v01.json).

| Slice | Records | Precision | Recall | F1 |
|---|---|---|---|---|
| slice-01 | 223 | 0.890 | 0.882 | 0.886 |
| slice-02 | 228 | 0.872 | 0.858 | 0.865 |
| slice-03 | 249 | 0.889 | 0.871 | 0.880 |
| slice-04 | 183 | 0.871 | 0.847 | 0.859 |
| slice-05 | 216 | 0.888 | 0.860 | 0.874 |